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Clinically oriented automatic three-dimensional enamel segmentation via deep learning

Wenting Yu[1], Xinwen Wang[2] and Huifang Yang[3*]

Abstract

Background Establishing accurate, reliable, and convenient methods for enamel segmentation and analysis is crucial for effectively planning endodontic, orthodontic, and restorative treatments, as well as exploring the evolutionary patterns of mammals. However, no mature, non-destructive method currently exists in clinical dentistry to quickly, accurately, and comprehensively assess the integrity and thickness of enamel chair-side. This study aims to develop a deep learning work, 2.5D Attention U-Net, trained on small sample datasets, for the automatical, efficient, and accurate segmentation of enamel across all teeth in clinical settings.

Methods We propose a fully automated computer-aided enamel segmentation model based on an instance segmentation network, 2.5D Attention U-Net. After data annotation and augmentation, the model is trained using manually annotated segmented enamel data, and its performance is evaluated using the Dice similarity coefficient metrics. A satisfactory image segmentation model is applied to generate a 3D enamel model for each tooth and to calculate the thickness value of individual enclosed 3D enamel meshes using a normal ray-tracing directional method.

Results The model achieves the Dice score on the enamel segmentation task of 96.6%. This study provides an intuitive visualization of irregular enamel morphology and a quantitative analysis of three-dimensional enamel thickness variations. The results indicate that enamel is thickest at the incisal edges of anterior teeth and the cusps of posterior teeth, thinning towards the roots. For posterior teeth, the enamel is thinnest at the central fossae area, with mandibular molars having thicker enamel in the central fossae compared to maxillary molars. The average enamel thickness of maxillary incisors, canines, and premolars is greater than that of mandibular incisors, while the opposite is true for molars. Although there are individual variations in enamel thickness, the average enamel thickness graduallly increases from the incisors to the molars among all teeth within the same quadrant.

Conclusions This study introduces an automatic, efficient, and accurate 2.5D Attention U-Net system to enhance precise and efficient chair-side diagnosis and treatment of enamel-related diseases in clinical settings, marking a significant advancement in automated diagnostics for enamel-related conditions.

Keywords Deep learning, Radiography, Enamel, 3D visualization, Artificial intelligence, Computer vision

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Background

Enamel is a unique tissue covering the dentin within the crown of teeth, appearing milky white, slightly transparent, and extremely hard, capable of withstanding strong chewing forces. Enamel is composed of approximately 98% inorganic material in the form of hydroxyapatite crystals, making it the hardest tissue in the human body [1, 2]. This chemical composition is a critical factor for dentists in determining treatment plans. The integrity of enamel is a key indicator for assessing the early stages of dental caries [3]. In orthodontic treatment, enamel reduction techniques between teeth are essential for addressing crowding, and the enamel surface undergoes structural changes with the bonding and removal of fixed orthodontic appliances or clear aligner attachments [4]. Accurate measurement of enamel thickness is also crucial for evaluating tooth wear and determining the need for subsequent restorative treatments in dentistry [5]. Moreover, the proportion of crown tissues and distribution of enamel thickness are reliable features used to infer taxonomic characteristics, phylogenetic relationships, dietary habits, and adaptive behaviors of fossils and modern humans, making them significant study subjects in anthropology and forensic science [6]. Therefore, establishing accurate, reliable, and convenient methods for segmenting and analyzing enamel is crucial for effectively planning endodontic, orthodontic, and restorative treatments, and for exploring mammalian evolutionary patterns.

Currently, traditional assessment methods for enamel integrity and thickness, such as manual measurement, physical sectioning, and ultrasonic measurement [7–9]. These methods are primarily used for quantifying enamel thickness in vitro but cannot measure enamel dimensions in vivo. They also have significant limitations in accurately describing irregular shapes. More importantly, traditional methods of assessing enamel on the two-dimensional level only obtain two-dimensional information on a certain cross-section. Consequently, the data obtained has significant limitations. In the past two decades, cone-beam computed tomography (CBCT) combined with 3D reconstruction software like 3-Matic and Mimics has replaced traditional geometric measurement methods [10]. Many systematic reviews support using CBCT to detect alveolar bone loss, particularly bifurcation lesions. CBCT's high accuracy has been demonstrated in evaluating geometric objects, dental defects, and alveolar bone defects, partly meeting the quantitative analysis requirements of clinical oral diagnosis and treatment planning [11].

However, without clinical application systems, segmentation of enamel structure based on CBCT can be challenging and labor-intensive. The irregular and complex structural shapes of enamel, coupled with limited clinical time, often lead oral healthcare providers to rely on visual inspection to assess enamel integrity and thickness changes [4]. This hasty approach can easily lead to incomplete anatomical information of enamel, affecting diagnosis, optimal treatment planning, or treatment failure. For example, early dental caries often involve enamel or even dentin damage (while patients usually do not have obvious bodily sensations), and the patient is easy to obtain CBCT data during routine oral examination. While in busy clinical work, a doctor can hardly spend time accurately measuring the enamel thickness of each tooth and determining its integrity, which could miss the best time for blocking early dental caries.

Accurate three-dimensional segmentation and quantitative analysis of enamel are the prerequisites and foundation for determining the condition of the enamel and subsequently taking appropriate treatment measures. Whether purely manual or semi-automatic segmentation programs, the structure of interest needs to be described through human observation. This requirement leads to subjective assessments by different examiners and may lead to observer fatigue, affecting the reliability of the technique [12]. Segmentation methods based on manually defined features have been widely studied in the past decade, including level sets, graph cuts, and template fitting [13–15]. However, these methods are not accurate in segmentation and require complex human initial intervention and correction; their learning ability is limited, highly sensitive to noise and complex situations, and low robustness in clinical environment processing. In recent years, deep learning (DL) based on convolutional neural networks (CNN) has made significant progress in automatic tooth segmentation due to its excellent image-processing capabilities. In the past five years, based on image classification networks, object detection, semantic segmentation, and instance segmentation in the dental field, including root, alveolar bone loss, and other automatic segmentation have performed well [16–18]. However, intelligent precision of irregular enamel structure segmentation and quantitative analysis still presents a challenge.

This study aims to develop a deep learning network based on small sample training to achieve three-dimensional automatic, efficient, and accurate segmentation of enamel based on CBCT images in clinical settings. It visually presents the irregular shape of enamel and quantitatively analyzes the changes in three-dimensional enamel thickness, thereby facilitating dentists to accurately and quickly determine the condition of the enamel, promoting chair-side diagnosis and treatment of enamel-related diseases.

Materials and methods

Data acquisition and image processing

The CBCT image data was selected from the outpatient database of Peking University School and Hospital of Stomatology. The female patient aged 26 underwent CBCT scanning using a CBCT machine i-CAT (Imaging Sciences International, Hatfield, PA, USA) at 120kVp, 18.45mAs, 20-second acquisition time, and 16×13 cm field of view. Voxel size is $250 \times 250 \times 250$ μm. The data were stored in the Digital Imaging and Communications in Medicine (DICOM) 3.0 format, following the guidelines for medical image data. The Hounsfield Unit (HU) values in CBCT data, representing gray scales, were normalized to a range of 0-1 for consistency and ease of analysis.

Data annotation and data augmentation

Data annotation

The CBCT image DICOM data sets of 30 teeth were imported into the image processing and 3D reconstruction software Dragonfly (v. 2024.1, Objects Research Systems, Montreal, Canada) for preprocessing and annotation. The process of image annotation and training neural networks can be conducted in the segmentation wizard module of the Dragonfly software. Due to issues such as poor enamel segmentation results on the occlusal surface previously, the coronal plane was chosen for image annotation and segmentation. The methodology of this study is depicted in Fig. 1.

The data processing in this study focuses on the enamel region of the teeth. We attempted to annotate and train the enamel using three different data generated from CBCT in three different directions. The data before and after annotation are shown in Fig. 1. To acquire gold-standard labels, every selected Frame was manually annotated into six different classes (enamel, dentin, cortical bone, trabecular bone, and pulp). "Frame" is defined as a single layer slice or a portion of this layer slice. Manual annotation was finished by an engineer and confirmed by an oral and maxillofacial radiologist.

Data augmentation

To expand the volume and diversity of image data, we developed a data augmentation strategy that employs targeted transformations with adversarial training instances. This method enriches the training dataset, enabling the model to generate a broader range of inputs. Data augmentation techniques were applied to all frames in the dataset, resulting in a tenfold increase in the quantity of annotated slices. The data augmentation parameters used during training in this study were set as follows: Data Augmentation times = 10, which involved applying horizontal flipping, vertical flipping, rotation up to 180 degrees, shearing up to 2 degrees, scaling between 90% and 110%, and adjusting brightness between 0.75 and 1.25. The dataset comprised 25,481,018 voxels; voxels used were 326,519(1.28%), of which 8,393 (2.57%) were marked as enamel. The total number of training patches is 5,615, and the total number of validation patches is 72. To develop and evaluate the model, we randomly partitioned our institutional dataset into training and validation subsets, with 80% of the data allocated for training and 20% for validation purposes. The validation subset was used exclusively to assess the model's performance.

Neural network

2.5D U-Net

The U-Net neural network, a convolutional neural network, derives its name from its network architecture's resemblance to the letter "U." The network comprises two primary components: the encoding and decoding layers, which enable image segmentation across any size.

Given the superior performance of Attention U-Net in small sample training and the benefits of U-Net for three-dimensional model reconstruction, we employ a 2.5D Attention U-Net approach (using the current layer and adjacent layers) for data training. This method aims to progressively expand the dataset to achieve an optimal neural network model for enamel segmentation.

The dataset was divided into two subsets, with 80% allocated for training and 20% for validation, to effectively monitor model performance and reduce the risk of overfitting during the training process. Additionally, an independent validation set comprising manually annotated regions with diverse anatomical variations was utilized. This external validation step was designed to assess the model's robustness and generalizability, ensuring that its performance was not overly optimized for the training data. The Attention U-Net model was provided by Dragonfly software, with input slices count set to 3, depth level to 4, and an initial filter count of 64. We employed Early Stopping techniques to prevent overfitting, which halted training when the Dice coefficient did not decrease over ten consecutive epochs. The resulting model, consisting of 5,528,393 parameters, was implemented within the Dragonfly on a Windows workstation, leveraging an NVIDIA Quadro RTX 5000 GPU for optimization.

Model evaluation

To quantify the 2.5D Attention U-Net network segmentation accuracy, the Dice similarity coefficient [19] was used. Surface deviations between the gold standards and the 2.5D Attention U-Net network-based 3D models were calculated to evaluate the segmentation accuracy around the edges of enamel structures.

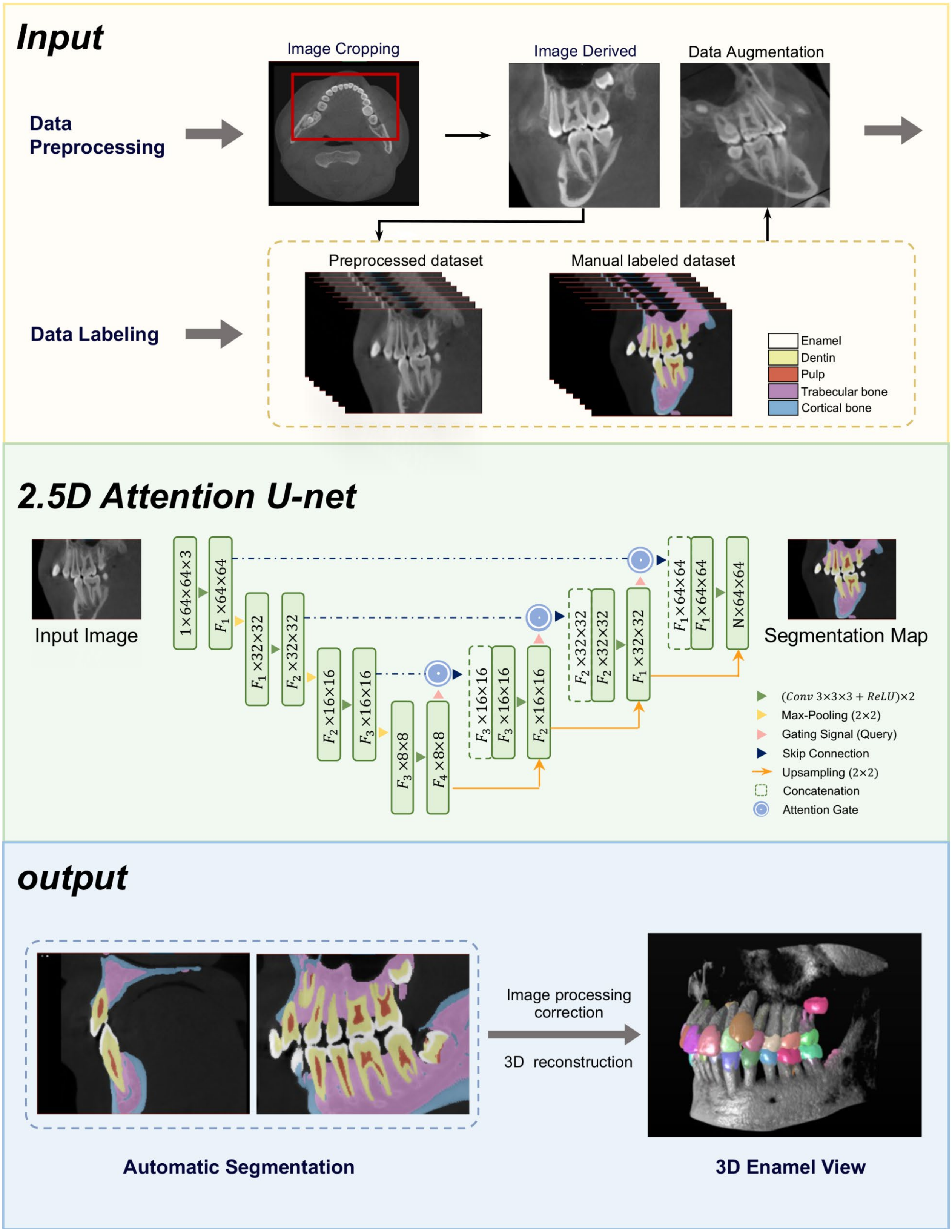

Fig. 1 Overview of the study recruitment process, gold standard generation, and network architecture of 2.5D Attention U-Net

Generation of the 3D surface model of enamel

Upon obtaining a satisfactory image segmentation model, the model will be applied to all sections to generate a comprehensive 3D segmentation model. Utilizing the distance map and watershed algorithm, teeth will be segmented from the entire region of interest (ROI) into multiple ROIs, with each tooth representing an individual ROI. Subsequently, Boolean operations will be employed to separate the enamel of each tooth into distinct ROIs, facilitating the creation of a 3D enamel model for each tooth. The thickness mesh tool will then generate thickness grids, which will undergo Laplacian smoothing with a single iteration to produce the final 3D enamel thickness model.

We computed thickness values from the mesh of a single, closed 3D shape using a directional method known as normal ray tracing. This method normally passes a ray to the mesh surface, and its intersection with the opposite surface is computed to obtain the thickness value. The Euclidean (straight line) distance between the intersections is considered the thickness at the given point. If the two surfaces containing the intersection points are not parallel, then the thickness values computed at the two opposite points will differ depending on the ray direction. This may lead to conflicting results for the pair of points.

The enamel segmentation was performed on CBCT images using the trained deep learning model, accurately delineating enamel regions as ROIs. For each tooth, the enamel volume was calculated by determining the total number of voxels within the segmented ROI and multiplying this by the voxel volume, derived from the known CBCT voxel dimensions. The specific calculation method is as follows:

$$V = N \times v$$

The enamel volume (V) was calculated by multiplying the total number of voxels (N) within the segmented region of interest (ROI) by the volume of a single voxel (v), where v is determined by the cube of the voxel size.

Results

The Dice Similarity Coefficient (DSC) of the segmentation results on the training dataset achieved a value of 96.0%. For enamel, the DSC was 96.6%. Using the trained model, we segmented the entire CBCT dataset, comprising 528 sagittal slices, to identify all seven regions of interest (ROIs). The segmented results of enamel are illustrated in Fig. 2.

Subsequently, each ROI was smoothed individually. The smoothed ROIs were then exported as STL files for further analysis. Two smoothing methods were considered: Laplacian Smoothing and Hamming Windowed Smoothing, to ensure high-quality surface reconstruction.

The 3D visualization of the segmentation and reconstruction outcomes of the patient's enamel is depicted in Fig. 3. The results showed the unsegmented original CBCT image, segmented enamel structure from CBCT based on the 2.5D Attention U-Net segmentation model, and three-dimensional morphology of the segmented enamel structure, respectively.

The segmentation results, morphological analysis, and the 3-dimensional reconstruction images for analyzing the enamel thickness are shown in Figs. 4, 5. The right maxillary first molar (Tooth 16), left maxillary central incisor (Tooth 21), left mandibular first molar (Tooth 36), and right mandibular central incisor (Tooth 41) were selected as representative teeth to detail the changes in enamel thickness.

At the two-dimensional level, the enamel thickness of each representative tooth was presented in spectral colors from both sagittal view, coronal view, and occlusal view (Fig. 4). Both maxillary and mandibular teeth exhibit the same vertical pattern of enamel thickness variation, the enamel thickness is greatest at the incisal edge of the anterior teeth and the cusp of the posterior teeth, with a decreasing trend towards the root. Notably, the enamel thickness on the lingual surface of tooth 21 first decreases, then increases and decreases again. The area where the enamel thickens on the lingual surface corresponds to the position of the cingulum (the contact point on the lingual surface of the upper and lower anterior teeth). For the posterior teeth occlusal surface, the enamel thickness is greatest at the distolingual cusp of tooth 16 and the distobuccal cusp of tooth 36, and thinnest at the central fossa area. However, the enamel thickness in the central fossa of the mandibular molar is greater than that in the maxillary molars.

The three-dimensional reconstructed images exhibit similar trends of enamel thickness variation to those in two-dimensional images (Fig. 5). However, the 3D images are more intuitive and make it easier to locate areas of different enamel thickness precisely. For example, in Fig. 5A, the enamel in the central fossa area is the thinnest, with a thickness of about 2.22 mm, and the range of the thinnest enamel area can be directly located.

We conduct a comprehensive statistical analysis to characterize the enamel thickness over all teeth in the oral cavity (Fig. 6). The results indicate that the average enamel thickness of the maxillary incisors, canines, and premolars is greater than that of their mandibular counterparts, while the opposite is true for molars. For all teeth in the same quadrant, the average enamel thickness gradually increases from the incisors to the molars. The minimum enamel thickness for all teeth in the oral cavity is 1.20 mm, and the maximum is 3.45 mm. The median average thickness of enamel for anterior teeth are central incisors (1.96 mm), lateral incisors (2.04 mm),

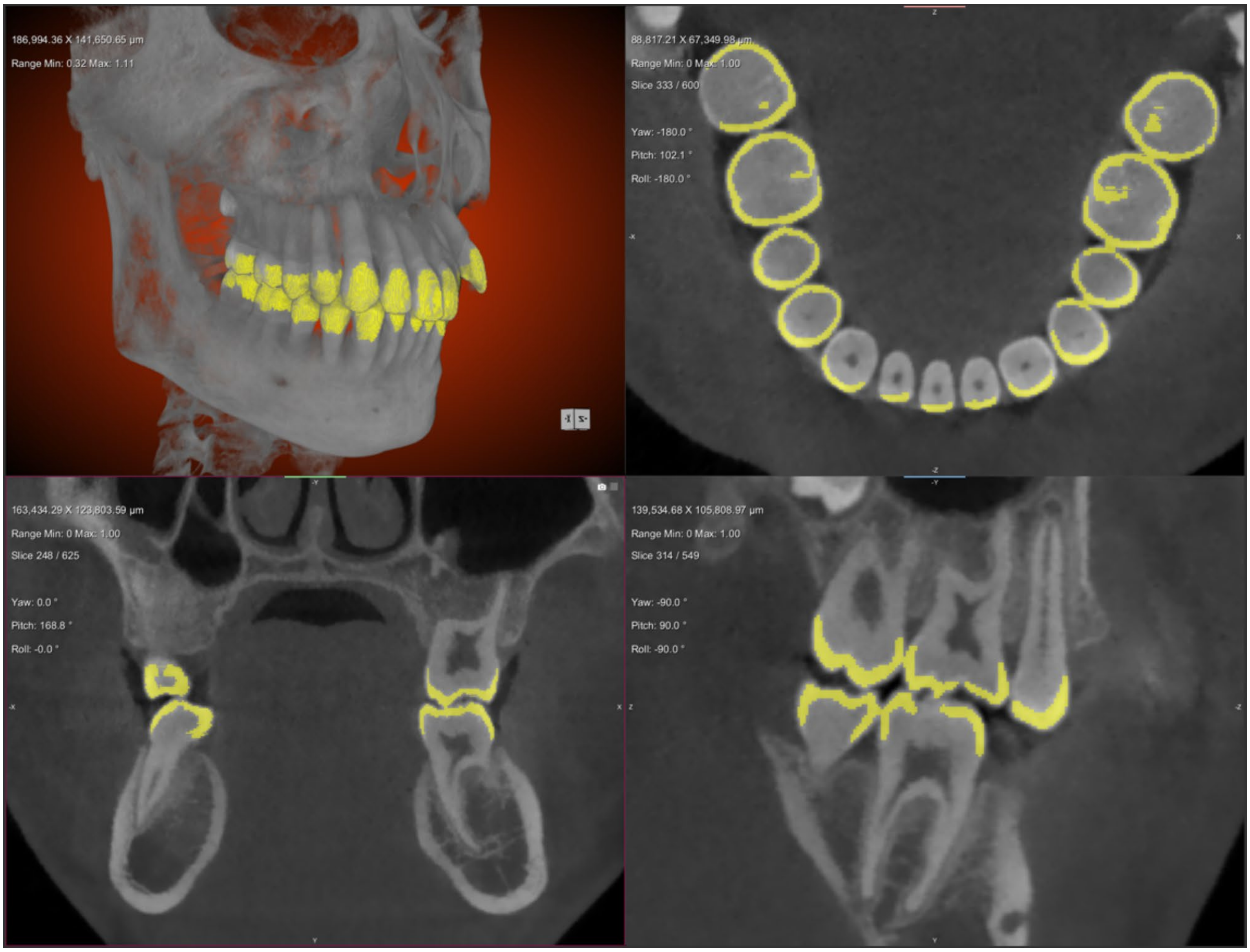

Fig. 2 The segmented images of enamel

and canines (2.24 mm). The average median thickness for premolars is 2.44 mm, and for molars, it is 2.59 mm.

We also conducted a detailed analysis of the volumetric data of all teeth (Table 1). The results indicate that the variation patterns in enamel volume are consistent with those observed in enamel thickness. The mandibular first molar exhibits the largest enamel volume (317.8 mm^3), whereas the mandibular second lateral incisor shows the smallest enamel volume (64.1 mm^3).

Discussion

Although the thickness of enamel is only a few millimeters, as the hardest tissue on the outside of the crown, it serves as the 'first line of defense' protecting teeth from external damage. Any disease, external mechanical injury, or iatrogenic operation that breaches the enamel will lead to irreversible rapid progression of dental diseases and damage to the function and aesthetics of the teeth [20]. The integrity of enamel is usually assessed by its thickness. Enamel thickness is of great importance in multiple disciplines as follows:

Biological and anatomical significance

Thicker enamel generally indicates that teeth can better resist external mechanical damage and chemical erosion, possessing higher hardness and wear resistance, while thinner enamel means teeth are more susceptible to microbial and other external stimuli [21]. Additionally, enamel thickness is an essential indicator in the study of tooth classification and system evolution [22]. There can be significant differences in enamel thickness among different types of teeth or different species, which helps us understand the evolutionary history and biodiversity of teeth.

Clinical significance in dentistry

Measuring enamel thickness aids in diagnosing dental diseases (such as caries, enamel hypoplasia, tooth wear, etc.), determining the maximum amount of interproximal enamel removal in orthodontics, and evaluating the preparation amount for ultra-thin veneers in restorative dentistry, guiding the limits of invasive dental treatments. Furthermore, understanding changes in enamel thickness

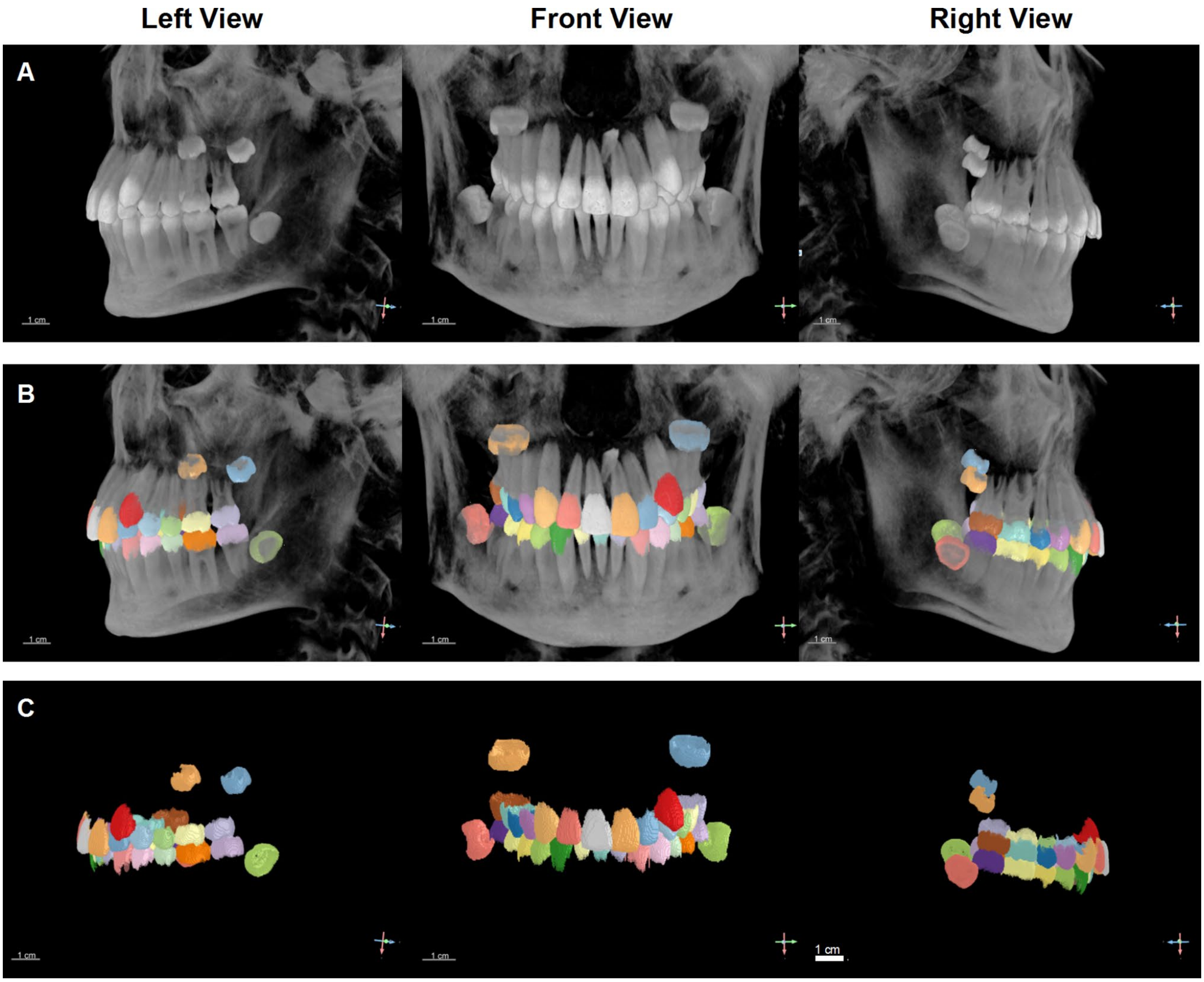

Fig. 3 The 3D visualization of individual segmented enamel. (**A**) Original input CBCT Image. (**B**) Segmented enamel structure from CBCT based on the 2.5D Attention U-Net segmentation model. (**C**) Three-dimensional morphology of the segmented enamel

helps guide patients in preventive care for teeth with thinner enamel [23].

Ecological significance

Enamel thickness is closely related to animal dietary habits [6]. Generally, animals that eat hard food have thicker enamel to withstand the immense pressure generated during chewing. Therefore, measuring enamel thickness can infer animal dietary habits and ecological niches.

In summary, the significance of enamel thickness spans biology, ecology, anatomy, clinical medicine, and dentistry. It is not only an essential indicator of oral health and function but also a crucial clue in studying biodiversity and biological evolution.

However, there is currently no mature, non-destructive method in clinical dentistry that can quickly, accurately, and comprehensively assess the integrity and thickness of the entire enamel in patients at the chair-side. Traditional methods such as manual measurement, optical coherence tomography, and stereomicroscopy can only measure the enamel thickness of extracted teeth and cannot accurately and comprehensively describe the irregular structure of enamel [3, 7, 24]. Although the development of cone-beam computed tomography (CBCT) and 3D reconstruction software (e.g., 3-Matic and Mimics) has evaluated irregular structures possible, due to the technical challenges posed by the "shell-like" complex structure of enamel, few studies are conducting three-dimensional quantitative analysis of the enamel thickness using CBCT. Akli et al. used CBCT data to manually measure the possible variations in the enamel thickness of upper anterior teeth [25], while Salam et al. manually measured the enamel thickness of mandibular canines and first molars. Both studies only considered the correlation between age and changes in enamel thickness [26]. Al-Zahawi et al. also manually measured the enamel

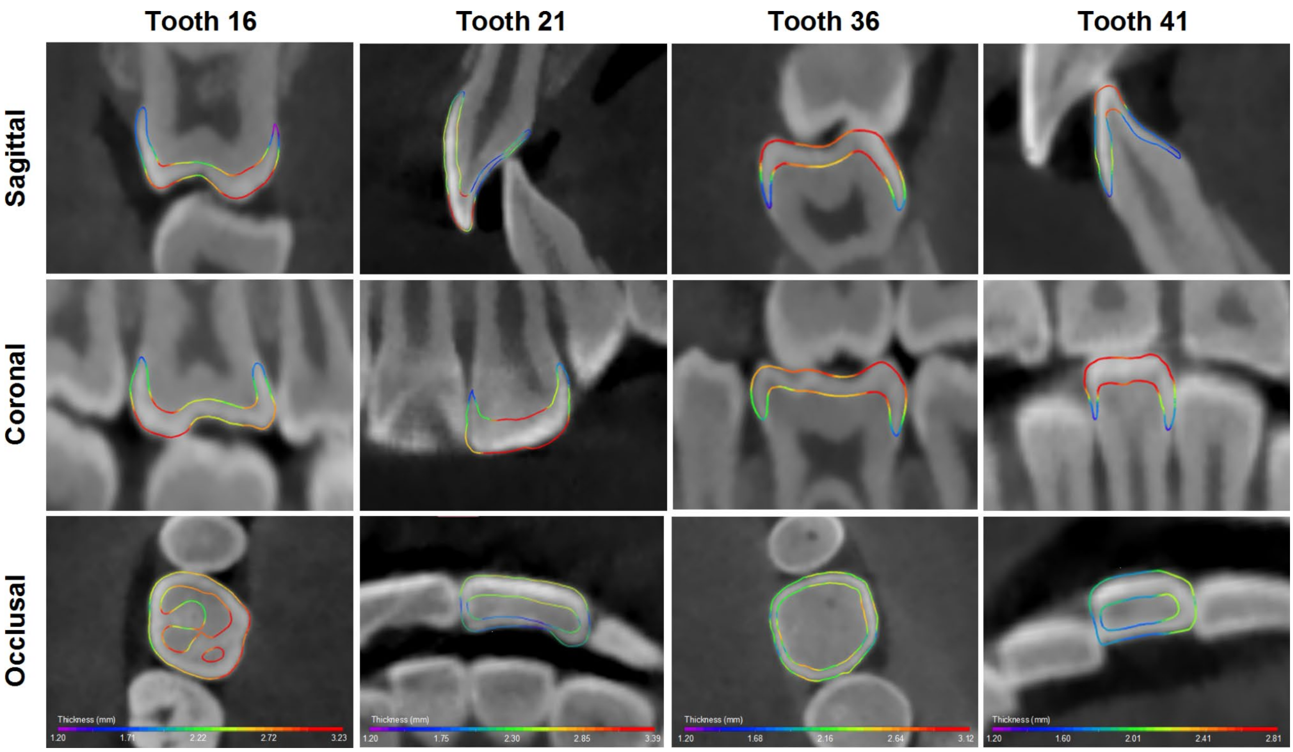

Fig. 4 Spectral map of two-dimensional distribution of enamel thickness. The right maxillary first molar (Tooth 16), left maxillary central incisor (Tooth 21), left mandibular first molar (Tooth 36), and right mandibular central incisor (Tooth 41) were selected as representative teeth to show the physiological variation trends in enamel thickness

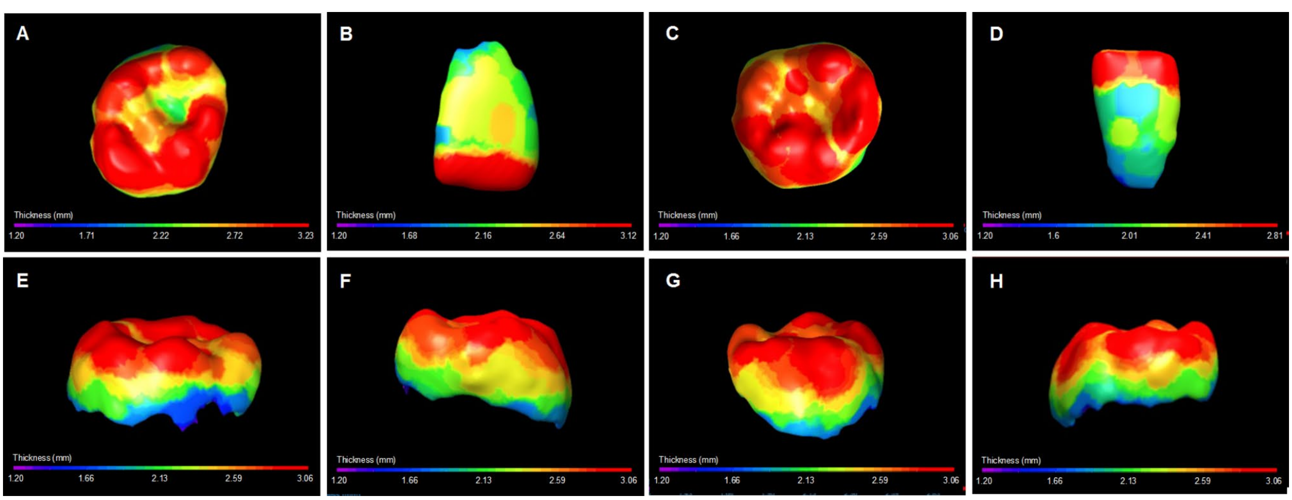

Fig. 5 Three-dimensional visualization and quantitative analysis of enamel thickness. (**A**) Occlusal surface of tooth 16; (**B**) Labial surface of 21; (**C**) Occlusal surface of tooth 36; (**D**) Labial surface of tooth 41; (**E**-**H**) Buccal surface, mesial surface, lingual surface, and distal surface of tooth 36

thickness of maxillary incisors and considered the correlation between gender and age with changes in enamel thickness [5]. However, these studies were based on researchers manually measuring the enamel thickness at certain cross-sections. Due to the irregular shape of enamel structures, the entire process is highly challenging, labor-intensive, and time-consuming and requires a high level of expertise from the researchers.

In recent years, deep learning (DL) based on convolutional neural networks (CNN) has made significant progress in the field of automatic tooth segmentation due to its excellent image processing capabilities, achieving an average Dice score of up to 95.1% for the teeth and maxillofacial bones segmentation [18]. However, few empirical studies exist on the intelligent, accurate segmentation of enamel structures. Tan et al. developed a multi-stage

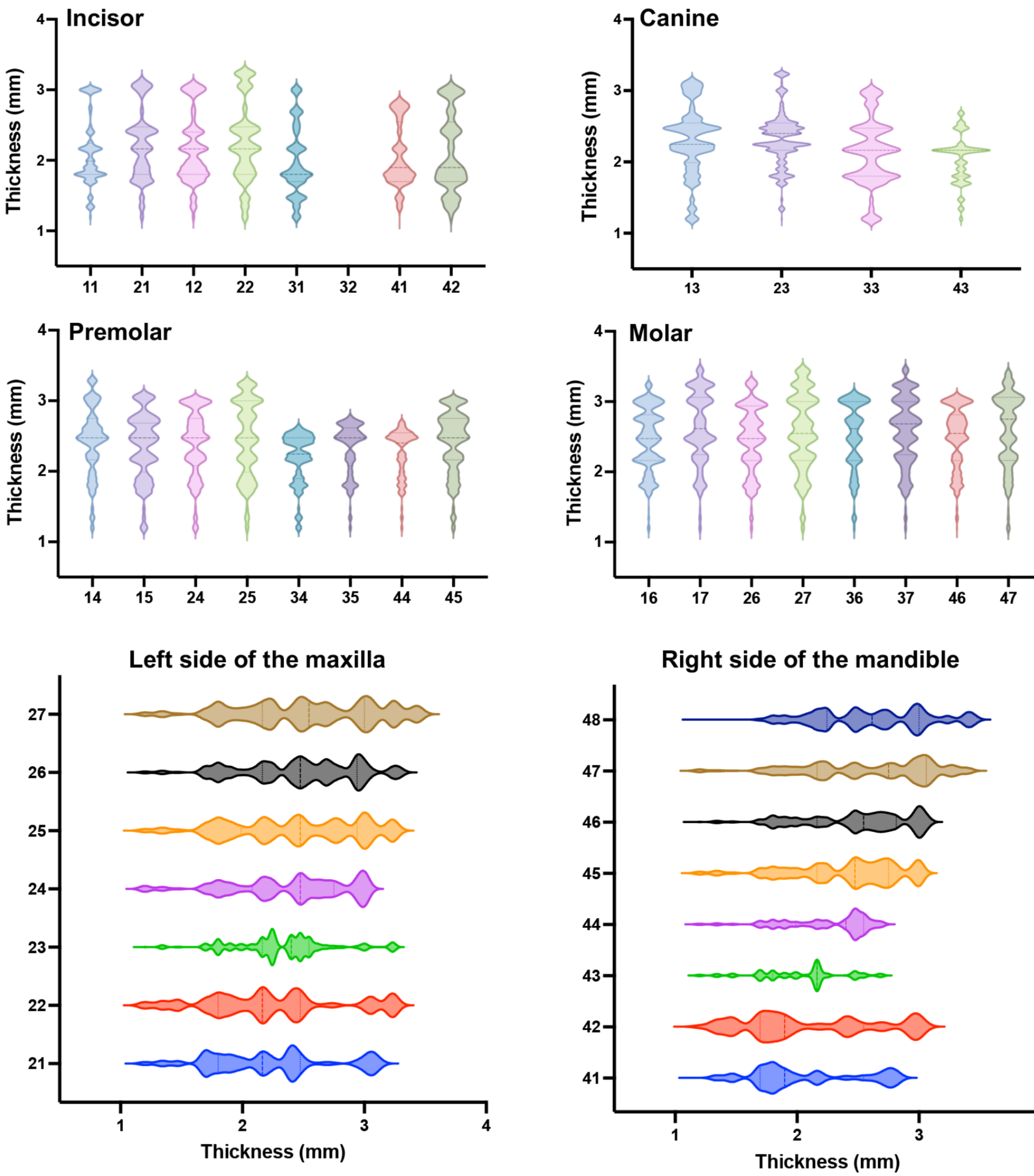

Fig. 6 Violin plots depicted the distribution of enamel thickness for each tooth. On average, the enamel thickness of the upper incisors, canines, and premolars exceeds that of their lower counterparts, whereas the reverse is true for molars. Within the same quadrant, enamel thickness consistently increases from the incisors to the molars for all teeth. (The patient is congenitally missing the right lateral incisors in the mandible)

Table 1 The distribution of enamel volume for each tooth (mm^3)

	Central incisor	Lateral incisor	Canine	First premolar	Second premolar	First molar	Second molar	Third molar
Left upper	172.9	119.4	197.9	202.4	176.7	298.1	321.2	/
Left Lower	72.7	/	130.0	140.5	143.7	317.8	303.5	266.2
Right upper	147.6	108.2	165.2	197.4	158.2	299.9	314.8	/
Right Lower	68.5	64.1	118.0	150.7	155.2	309.0	301.4	287.7

convolutional neural network framework to finely segment tooth substructures (including enamel) [17]. However, it did not focus on enamel thickness information, and the training of the segmentation model was based on large samples, making the data preprocessing work very complex and increasing the difficulty of practical application.

In this study, we developed an adaptive algorithm based on semantic segmentation, which improved the efficiency of the segmentation model through data preprocessing, reducing the required sample size to single digits [27]. Experimental results indicated that the Dice score for the enamel segmentation task was 96.6%, achieving accurate segmentation of enamel. To the best of our knowledge, this is the first study to use deep learning methods to automatically segment enamel for all teeth in the oral cavity and achieve single-tooth enamel thickness measurement and three-dimensional visualization. This network can achieve model training and automatic segmentation based on a single patient's CBCT data, which is crucial because it allows for rapid chair-side 3D visualization of the full-mouth enamel and its thickness during clinical treatment planning. Thereby enables the dentist to identify specific teeth more prone to disease for a particular patient, facilitating a precise, rapid, and efficient clinical workflow.

The enamel segmentation and 3D visualization based on the 2.5D Attention U-Net we proposed have broad potential clinical applications in the field of dentistry. In the field of endodontics, it can be used for early assessment of caries progression to reduce severe structural and functional damage to teeth caused by hidden caries. In orthodontics, chair-side precise, personalized, and rapid three-dimensional enamel thickness visualization assessment is vital for determining the boundaries of interproximal enamel reduction and enameloplasty procedures and for conducting orthodontic treatment based on the premise of tooth health. In prosthodontics, precise assessment of enamel thickness is the cornerstone of aesthetic restoration with minimal tooth preparation design.

However, our study still has some limitations. For example, the included dataset does not cover patients with metal artifacts or those already wearing metal braces, nor does it include primary teeth with different degrees of mineralization compared to permanent teeth. Therefore, further exploration is needed to assess the applicability of this method to a broader population and various clinical scenarios.

Conclusion

This study proposed a deep learning network based on small sample training to achieve automatic, efficient, and accurate 3D enamel segmentation in clinical settings. It visually presents the irregular shape of enamel and quantitatively analyzes the changes in three-dimensional enamel thickness, thereby facilitating dentists to accurately and quickly determine the condition of the enamel, promoting chair-side diagnosis and treatment of enamel-related diseases.

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Author contributions
YWT contributed to investigation, validation, visualization, writing – original draft, writing – review & editing, and funding acquisition. WXW contributed to resources, data curation, conceptualization. YHF contributed to methodology, software, project administration, Writing – review & editing and Funding acquisition. All authors gave final approval and agree to be accountable for all aspects of the work.

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This retrospective study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Biomedical Ethics Committee of Peking University School and Hospital of Stomatology (No. PKUSSIRB – 202495005). Given that this research is a retrospective study, the Ethics Committee waived the requirement for informed consent. All the methods in the study were carried out following the relevant guidelines and regulations.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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